

Java Full Stack Internship – Task 3

Task Description

In Task-2, you connected your contact form to **MySQL** and saved submissions in a database. Now, let's move ahead and create a **Contact Management Dashboard** where you can **view all submitted contacts in a web page**.

This will introduce you to:

- Fetching data from backend with Spring Boot.
- Returning JSON data from REST API.
- Displaying dynamic data on frontend using HTML + JavaScript.

What You'll Build (Requirements)

1. Backend (Spring Boot):

- Already saving contacts → Now add an endpoint `/contacts` that returns all contacts as JSON.

2. Frontend (HTML + JS):

- Create a **Contacts Page**.
- Use **JavaScript fetch API** to call `/contacts`.
- Display results in a **table format**.

Deliverable:

A contact management system where users can submit data (form) and an admin can **view all saved contacts in a table**.

Why This Task Is Important

- Introduces **REST API concepts**.
- Helps you practice **fetching JSON data** and displaying it.
- Bridges the gap between frontend and backend.

Sample Starter Code (Simplified & Explained)

Spring Boot – Controller

java

```
@GetMapping("/contacts")
public List<Contact> getAllContacts() {
    return repo.findAll(); // returns JSON list of contacts
}
```

Frontend – contacts.html

html

```
<!DOCTYPE html>
<html>
<head>
    <title>Contacts</title>
</head>
<body>
    <h2>All Contacts</h2>
    <table border="1">
        <thead>
            <tr><th>ID</th><th>Name</th><th>Email</th><th>Message</th></tr>
        </thead>
        <tbody id="contactTable"></tbody>
    </table>

    <script>
        fetch("http://localhost:8080/contacts")
        .then(res => res.json())
        .then(data => {
            let table = document.getElementById("contactTable");
            data.forEach(c => {
                let row = `<tr>
                    <td>${c.id}</td>
                    <td>${c.name}</td>
                    <td>${c.email}</td>
                    <td>${c.message}</td>
                </tr>`;
                table.innerHTML += row;
            });
        });
    </script>
</body>
</html>
```

Free Video Courses / Tutorials

- [Spring Boot Crash Course – Amigoscode](#)
- [Spring Boot + MySQL Integration](#)
- [Spring Boot REST API Tutorial](#)

Documentation

- [Spring Boot Docs](#)
- [Spring Data JPA Docs](#)
- [Baeldung Spring Boot Tutorials](#)

Tools

- [Spring Initializr](#) → Generate projects
- [MySQL Workbench](#)
- [IntelliJ IDEA Community](#) / [VS Code](#)